

list1=[2,3,4,4,4,5,5,5,6,6,6,6,7,7,7,7,8,9]
target=6

Step 1

Initialize: low=0, high=17, first=-1

Step 2

Iteration 1: mid=(0+17)//2=8, list1[8]=6. Match found. first=8, high=7.

Step 3

Iteration 2: low=0, high=7, mid=3, list1[3]=4. 6>4 so low=4.

Step 4

Iteration 3: low=4, high=7, mid=5, list1[5]=5. 6>5 so low=6.

Step 5

Iteration 4: low=6, high=7, mid=6, list1[6]=5. 6>5 so low=7.

Step 6

Iteration 5: low=7, high=7, mid=7, list1[7]=5. 6>5 so low=8.

Step 7

Exit: low=8, high=7 so loop stops.

Step 8

Since first=8 (not -1), output: The first occurrence of the target is at 8.

Iteration	low	high	mid	Action	first
Start	0	17	-	Initialize	-1
1	0	17	8	Found, high=7	8
2	0	7	3	low=4	8
3	4	7	5	low=6	8
4	6	7	6	low=7	8
5	7	7	7	low=8	8
Exit	8	7	-	Loop ends	8